

Ethan Chang

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EDUCATION

UNIVERSITY OF CALIFORNIA, BERKELEY

Berkeley, CA | August 2022 - May 2026

Bachelor's of Art in Data Science & Minor in Computer Science

Relevant Coursework: Artificial Intelligence, Linux Systems, Probability Theory & Principles of Data Science, Discrete Mathematics & Probability Theory, Linear Algebra & Differential Equations, Data Structures & Algorithms, Structure & Interpretation of Computer Programs

RELEVANT EXPERIENCE

Berkeley Discovery | Database Assistant

February 2024 - Present

- Analyzed academic/professional listings for high-priority posts and updated the database with weekly opportunities for over **6500+** concurrent users, boosting overall database activity by **50%**.
- Routinely manage **1200+** active listings and identify pain points on data infrastructure quality to improve the Discovery user interface with system modifications, design updates, and increased user bases.

Innovative Design at Berkeley | Frontend Developer & Web Designer

September 2023 - May 2024

- Designed mock-up graphic **UX** redesigns of the Berkeley Undergraduate website and designed a mobile application version, prioritizing intuitive and optimal user interfaces.
- Leveraged **HTML** and **CSS** through **React.js** to translate team-based high fidelity designs into functional and aesthetic web pages for the client through strong client-team collaboration.

Taiwanese American Student Association | Web Officer

December 2022 - December 2023

- Troubleshooted **Linux** system issues and routinely cleaned/updated database data through **SQL** management.
- Programmed interactive website features and maintained website content using **HTML**, **CSS**, and **JavaScript**.
- Monitored website traffic and user analytics to drive scalability and functionality to match the organization's focus.

PROJECTS

Neural Network Modeling

November 2024

- Developed binary perceptron, language identification, and convolutional neural network models using **PyTorch**, **numPy**, and **Matplotlib** through **Python** to solve machine learning and classification problems.
- Trained a regression model to approximate a Sine wave using a ReLU activation function and a digit classification model to classify handwritten digits from the MNIST dataset, producing a **98%** model accuracy.

Outreach Effectiveness Data Analysis | Berkeley Discovery

July 2024 - August 2024

- Employed data functions via **Pandas** in JupyterLab to analyze user login frequency and activity from database datasets informing data-driven decisions about the initiative's outreach funding allocation and event effectiveness.
- Used EDA through **data modeling** and quantitative insights to identify key trends/patterns between time periods, visualizing a near **200%** boost in user-database activity from in-person initiatives.
- Standardized data analysis methodologies for administrative usage regarding future outreach effort and event datasets.

2D-World Interactive Game

April 2023

- Programmed a 2D tile-based world engine and developed an interactive user game using **Java** data algorithms and packages with player functionality.
- Integrated an engaging **user interface** with unique game features including NPC/item encounters with combat screens, level progression, and player data tracking.

LEADERSHIP

Ighota! Virtual Learning | Vice President

June 2021 - July 2024

- Researched STEM curriculum for lab worksheets and designed 15+ graphic review sheets for **600+** Nigerian students.
- Coordinated with the Girl's Secondary School in Nigeria, providing **20+** secondary schools with our STEM resources.
- Outreached to several Nigerian institutions, garnering **8+** active educators to record STEM material for our students.

Taiwanese American Student Association | Internal Vice President

December 2023 - May 2024

- Successfully organized 5 general events for **100+** students, promoting an environment to learn about Taiwanese culture.
- Organized the semesterly retreat, weekly team socials and meetings, and bonding activities for all cabinet members.
- Actively fostered camaraderie, cabinet bonding, and strong collaboration among **30+** cabinet members across **7** teams.

SKILLS & INTERESTS

- Skills & Knowledge:** Python, Java, SQL, HTML, CSS, JavaScript, React.js, Git, jQuery, Pandas, Sklearn, PyTorch
- Interests:** Fullstack, AI/ML, Data Analytics & Engineering, Web Development, Product Design, Sketching, Photography